

No. Laporan: KALX62509125100R

Sertifikat Kalibrasi Eksternal

Identitas Pemilik : Laboratorium Telkom Test House (TTH)
Jl. Gegerkalong Hilir No. 47 Bandung 40152

Deskripsi Alat yang Dikalibrasi

Nama Alat : ~12.6 km LENGTH OF SINGLE MODE OPTICAL FIBRE
Merek : CORNING
Tipe/Model : ~12.6 km
Nomor Seri : 100547391626
Identifikasi Alat : 010/KAL

Informasi Kalibrasi

Tanggal Diterima CSTTH : 07 November 2025
Tanggal Diterima Lab : 19 November 2025
Penyedia Kalibrasi : Lab Kalibrasi Telkom Jakarta (LK-016-IDN)
Tempat Kalibrasi : Lab Kalibrasi Telkom Jakarta
Tanggal Pengiriman : 20 November 2025
Tanggal Pengambilan : 20 November 2025
Nomor Sertifikat : JCL/25/3210/DY
Tanggal Kalibrasi : 20 November 2025
Tanggal Kalibrasi Ulang : 20 November 2026

Diperiksa Oleh:



ADI PERMADI

Mgr. Standardization & Suitability Assurance

Bandung, 14 Januari 2026

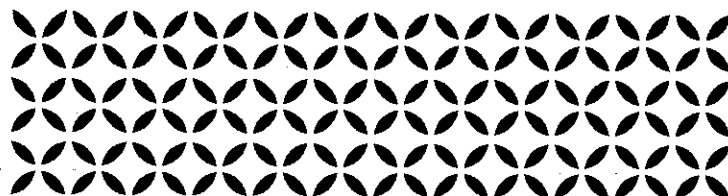
Disetujui Oleh:



GUNADI DWI HANTORO

SM Infrastructure Service Research & Assurance





No. Dokumen : Form TCL-QP-19-01
No. / Tgl. Terbit : 01 / 20 Juli 2017
No. / Tgl. Revisi : 02 / 04 Januari 2021



CERTIFICATE OF CALIBRATION

P T T E L K O M I N D O N E S I A T b k
KAN ACCREDITED CALIBRATION LABORATORY (LK-016-IDN)

Jakarta : Jl. Percetakan Negara No.17-19 Jakarta Pusat 10570 Phone 021-21480341
Medan : Jl. Gaharu No. 1 Medan 20235 Phone 061-4531311
Surabaya : Jl. Gayungan PTT No. 17-19 Surabaya 60235 Phone 031-8283678
Makassar : Jl. AP Pettarani No. 4 Makassar 90221 Phone 0411-861153

CS Support
☎ 081-16-4-17025

Certificate Number
JCL/25/3210/DY
Approved Signatory

M. Ilyas

Date of Issue : 21-Nov-2025
Submitted by : PT. TELKOM INDONESIA, Tbk.
Laboratorium Telkom Test House (TTH)
Attention : Mgr. Standardization & Suitability Assurance
Description : Optical Fibres
Manufacturer/Model : NPL / ~ 12.6 km
Serial Number : 100547391626 / (213/CAL)
Environment : 23.4 °C ± 0.2 °C 56.2 % ± 0.1 % RH
Calibration Location : In-House
Date Calibrated : 20-Nov-2025
Recommended Due Date : 19-Nov-2026
Date Received : 20-Nov-2025

PT. Telkom Indonesia, Tbk. - Telkom Certifies that this instrument has been calibrated under the environmental condition stated above using telkom standard (s) which is (are) traceable to the international system of Units (SI) through MIKROKOM. S.r.o Praha 4, Czech Republic.

Method of Calibration :

The calibration was carried out based on the established Standard Telkom Calibration Procedure, TCL-WI-0009 and by comparison against the following standard(s) and instrument(s) which is (are) calibrated on schedule to maintain traceability and required accuracy :

Description	S/N	Cal Date	Due Date
1. OTDR	08795	04-Sep-2024	03-Sep-2027

CERTIFICATE OF CALIBRATION

Checked By : 

Certificate Number : JCL/25/3210/PY

Setup

From end A



Measurements λ 1310 nm

TEST No.	EXPECTED VALUE OF UUT	MEASURED VALUE	CORRECTION	UNCERTAINTY
1.	13285.97 m	13285.32 m	0.65 m	$\pm$ 2.6 m

Measurements λ 1310 nm

Section (km)	Attenuation from end A	Attenuation from end B	Average attenuation measurement	Expected Attenuation	Correction
1.0 to 2.0	0.317 dB/km	0.317 dB/km	0.317 dB/km	0.317 dB/km	0.000 dB
2.0 to 3.0	0.313 dB/km	0.317 dB/km	0.315 dB/km	0.316 dB/km	0.001 dB
3.0 to 4.0	0.317 dB/km	0.317 dB/km	0.317 dB/km	0.317 dB/km	0.000 dB
4.0 to 5.0	0.315 dB/km	0.315 dB/km	0.315 dB/km	0.317 dB/km	0.002 dB
5.0 to 6.0	0.315 dB/km	0.316 dB/km	0.316 dB/km	0.316 dB/km	0.001 dB
6.0 to 7.0	0.317 dB/km	0.317 dB/km	0.317 dB/km	0.317 dB/km	0.000 dB
7.0 to 8.0	0.318 dB/km	0.319 dB/km	0.319 dB/km	0.318 dB/km	-0.001 dB
8.0 to 9.0	0.316 dB/km	0.316 dB/km	0.316 dB/km	0.318 dB/km	0.002 dB
9.0 to 10.0	0.316 dB/km	0.317 dB/km	0.317 dB/km	0.317 dB/km	0.001 dB
10.0 to 11.0	0.320 dB/km	0.316 dB/km	0.318 dB/km	0.318 dB/km	0.000 dB
11.0 to 12.0	0.318 dB/km	0.316 dB/km	0.317 dB/km	0.316 dB/km	-0.001 dB

Measurements λ 1550 nm

TEST No.	EXPECTED VALUE OF UUT	MEASURED VALUE	CORRECTION	UNCERTAINTY
1.	13291.44 m	13290.46 m	0.98 m	$\pm$ 2.7 m

Measurements λ 1550 nm

Section (km)	Attenuation from end A	Attenuation from end B	Average attenuation measurement	Expected Attenuation	Correction
1.0 to 2.0	0.178 dB/km	0.179 dB/km	0.179 dB/km	0.180 dB/km	0.001 dB
2.0 to 3.0	0.178 dB/km	0.176 dB/km	0.177 dB/km	0.178 dB/km	0.001 dB
3.0 to 4.0	0.179 dB/km	0.179 dB/km	0.179 dB/km	0.179 dB/km	0.000 dB
4.0 to 5.0	0.181 dB/km	0.179 dB/km	0.180 dB/km	0.179 dB/km	-0.001 dB
5.0 to 6.0	0.179 dB/km	0.178 dB/km	0.179 dB/km	0.179 dB/km	0.001 dB
6.0 to 7.0	0.181 dB/km	0.178 dB/km	0.180 dB/km	0.179 dB/km	-0.001 dB
7.0 to 8.0	0.178 dB/km	0.182 dB/km	0.180 dB/km	0.179 dB/km	-0.001 dB
8.0 to 9.0	0.182 dB/km	0.176 dB/km	0.179 dB/km	0.180 dB/km	0.001 dB
9.0 to 10.0	0.178 dB/km	0.179 dB/km	0.179 dB/km	0.178 dB/km	-0.001 dB
10.0 to 11.0	0.176 dB/km	0.175 dB/km	0.176 dB/km	0.179 dB/km	0.003 dB
11.0 to 12.0	0.178 dB/km	0.179 dB/km	0.179 dB/km	0.178 dB/km	-0.001 dB

ESTIMATED UNCERTAINTIES

Test	Estimated uncertainty
1 Length Test λ 1310 nm	$\pm$ 2.6 m
2 Length Test λ 1550 nm	$\pm$ 2.7 m
3 Attenuation λ 1310 nm	$\pm$ 0.096 dB
4 Attenuation λ 1550 nm	$\pm$ 0.094 dB